

A Project Report on
IMAGE FORGERY DETECTION USING HYBRID
CNN-SIAMESE NETWORKS

Submitted in partial fulfillment of the requirements for the award of the degree

of
BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted By

NAME	ROLL NO.
GOTTAM PUJITHA	(22NE1A0439)
DOREDLA BHUVAN KUMAR	(22NE1A0429)
BODDU DIVYA	(22NE1A0418)
GUNJI VENU	(22NE1A0440)

Under the Esteemed Guidance of

Mrs. G. Suseelamma, M.Tech.,(Ph.D).

Mrs. P. Venkata Ramana, M.Tech

Associate Professor

Assistant Professor

Department of Electronics and Communication Engineering



TIRUMALA ENGINEERING COLLEGE

(Autonomous)

An ISO 9001:2015 Certified Institution Accredited by NAAC (A+) & NBA
(APPROVED BY AICTE, NEW DELHI & AFFILIATED TO JNTUK,KAKINADA)

Jonnalagadda,Narasaraopet-522601,PALNADU(Dt.),A.P

2022-2026

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CERTIFICATE

This is to certify that the thesis entitled **“IMAGE FORGERY DETECTION USING HYBRID CNN-SIAMESE NETWORKS”** is a bonafide work done by **G. PUJITHA (22NE1A0439), D. BHUVAN KUMAR (22NE1A0429), B. DIVYA (22NE1A0418), G. VENU (22NE1A0440)** submitted in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in Electronics and Communication Engineering** from **Jawaharlal Nehru Technological University, Kakinada** during the academic year **2022–2026** under our guidance and supervision and worthy of acceptance as per the requirements of the university.

Internal Guide

Head of the Department

External Examiner

DECLARATION

We are the students of Tirumala Engineering College hereby declare that this project report entitled **“IMAGE FORGERY DETECTION USING CNN-SIAMESE NETWORKS”** being submitted to the Department of ECE, affiliated to JNTUK, Kakinada for the award of **“BACHELOR OF TECHNOLOGY”** in Electronics and Communication Engineering is a record of bonafide work done by us and it has not been submitted to any other Institute or University for the award of any other degree or prize.

Project Associates

GOTTAM PUJITHA	(22NE1A0439)
DOREDLA BHUVAN KUMAR	(22NE1A0429)
BODDU DIVYA	(22NE1A0418)
GUNJI VENU	(22NE1A0440)

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Project Associates

GOTTAM PUJITHA	(22NE1A0439)
DOREDLA BHUVAN KUMAR	(22NE1A0429)
BODDU DIVYA	(22NE1A0418)
GUNJI VENU	(22NE1A0440)

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ABSTARCT

Image forgery has become a serious issue due to the rapid advancement of digital image editing tools and artificial intelligence-based image generation technologies. Manipulated images can easily spread misinformation and create problems in areas such as journalism, legal investigations, social media, and digital forensics. Therefore, detecting forged images has become an important task in ensuring the authenticity and reliability of digital media. The main objective of this project is to develop an intelligent system that can detect whether an image is authentic or manipulated using deep learning techniques. In this project, a hybrid approach combining Convolutional Neural Networks (CNN) and Siamese Networks is used for effective image forgery detection. The system utilizes transfer learning with ResNet architectures such as ResNet-18, ResNet-50, and ResNet-101, which are pre-trained on the ImageNet dataset and fine-tuned for binary classification of authentic and forged images. A dataset containing both genuine and manipulated images is used for training and testing the models, and the images are preprocessed to ensure balanced and efficient learning. In addition to deep learning predictions, several image forensic techniques are applied to improve detection reliability, including noise level analysis, JPEG compression artifact detection, edge consistency analysis using the Canny edge detection algorithm, color distribution analysis, and local variance analysis for identifying abnormal texture patterns. These techniques contribute to a weighted forgery score that helps detect suspicious images even when the neural network prediction is uncertain. The performance of the models is evaluated using metrics such as accuracy, precision, recall, specificity, and F1-score. Experimental results show that deeper architectures such as ResNet-101 provide improved performance in detecting forged images. To make the system user-friendly, a web application is developed using the Flask framework that allows users to upload images and receive instant detection results. The developed system provides an effective and practical solution for image forgery detection and can assist journalists, forensic investigators, and cybersecurity professionals in verifying the authenticity of digital images.

CHAPTER-1

INTRODUCTION

1. INTRODUCTION

1.1 Introduction

The rapid advancement of digital technology has transformed the way images are created, shared, and consumed in modern society. Digital images have become an integral part of daily communication, news reporting, legal documentation, medical diagnosis, and social media interactions. People generally trust photographs as truthful representations of reality. However, the widespread availability of sophisticated image editing software and artificial intelligence tools has made it increasingly easy to manipulate digital images in realistic ways. Today, forged images can be created with such high quality that even trained experts may find it difficult to distinguish between authentic and manipulated content [1], [2].

Image forgery poses serious threats across multiple domains. In journalism and media, manipulated images can spread false information, influence public opinion, and damage the credibility of news organizations. Fake images may be used to misrepresent important events or create misleading narratives that affect political and social outcomes. In the legal system, altered images presented as evidence can compromise investigations and lead to incorrect judicial decisions. The medical field is also vulnerable, as manipulated medical images may result in incorrect diagnoses or unreliable research conclusions. Similarly, in financial sectors, forged identity images and documents are commonly used in fraud cases such as insurance scams and identity theft. On social media platforms, manipulated images spread rapidly and can create confusion, misinformation, and erosion of public trust [3], [4].

The concept of image manipulation is not new. Even in the early days of photography, images were modified using traditional darkroom techniques such as cropping, dodging, and burning. However, modern digital technology has made image editing significantly easier and more accessible to ordinary users. Software tools such as Adobe Photoshop, GIMP, and various mobile applications provide users with advanced editing capabilities that were once available only to professionals. Furthermore, recent developments in artificial intelligence, particularly deep learning and generative models, can now create highly realistic synthetic images that are nearly indistinguishable from real photographs. These technological advances have substantially increased the difficulty of identifying forged images using traditional methods [5], [6].

Given these challenges, there is a growing and urgent need for intelligent, automated systems that can accurately detect manipulated images. Developing reliable image forgery detection systems has emerged as an important research area within digital forensics and cybersecurity. Such systems can help verify the authenticity of digital images, prevent the spread of misinformation, and protect the integrity of visual evidence in various professional and legal

contexts. Automated detection systems offer the advantage of speed, consistency, and the ability to analyse large volumes of image data that would be impossible to examine manually [7], [8].

1.2 Applications of Image Forgery Detection System

The development of an intelligent image forgery detection system has several important applications across different sectors. Since manipulated images can cause misinformation, fraud, and security threats, an automated detection system can help verify the authenticity of digital images. By analysing image features and identifying suspicious patterns, the system provides valuable support in detecting forged or manipulated images. The following are some key applications of the proposed system [9].

1. Image Authentication

One of the primary applications of this system is verifying whether a digital image is authentic or manipulated. The system analyses various image characteristics and identifies possible signs of tampering. This helps ensure that images used in media, research, and documentation are genuine and trustworthy [10].

2. Journalism and Media Verification

The system can help journalists and media organizations verify the authenticity of images before publishing them. Since manipulated images can spread misinformation and influence public opinion, detecting forged images helps maintain the credibility and reliability of news content [11].

3. Digital Forensics and Criminal Investigation

Image forgery detection is very useful in digital forensic investigations. Law enforcement agencies can use the system to analyse photographic evidence and determine whether it has been altered or manipulated. This can help investigators identify fake evidence and support fair legal proceedings [1], [12].

4. Social Media Content Verification

On social media platforms, manipulated images often spread quickly and create confusion or misinformation. The proposed system can help detect forged images and prevent the spread of misleading visual content online [13].

5. Cybersecurity and Fraud Prevention

The system can also be used to detect fraudulent images used in identity verification, insurance claims, or financial scams. By identifying manipulated images, organizations can reduce fraud and improve security [14].

1.3 Area of the Project

The proposed project titled “Image Forgery Detection Using Deep Learning” belongs to multiple interdisciplinary domains that combine computer vision, artificial intelligence, and digital forensics. The system focuses on identifying manipulated images by analyzing visual patterns and statistical inconsistencies present within digital images. Since the project involves image analysis, machine learning algorithms, and digital media authentication, it covers several important technological and research areas [15].

1. Digital Image Forensics

The project belongs to the domain of digital image forensics, which focuses on verifying the authenticity of digital images and identifying whether an image has been manipulated. Many images can be altered using editing tools to hide information or mislead viewers. Digital image forensics analyzes image properties such as noise patterns, compression artifacts, and structural inconsistencies to detect such alterations. The proposed system uses these characteristics to determine whether an image is genuine or forged [1], [2].

2. Computer Vision

The project also falls under the field of Computer Vision, which involves enabling computers to interpret and analyze visual information from images and videos. Computer vision techniques allow machines to extract meaningful patterns and features from image data. In the proposed system, computer vision methods are used to process and analyze image pixels, detect irregularities, and identify patterns that may indicate image manipulation. By automatically analyzing visual content, the system can distinguish between authentic and tampered images, demonstrating the practical application of computer vision technology [7].

3. Artificial Intelligence

Artificial Intelligence plays a crucial role in the proposed system because it enables the development of intelligent models capable of detecting image forgery automatically. Traditional image analysis methods rely heavily on manual inspection, which is time-consuming and often unreliable. AI techniques allow the system to learn from large datasets of authentic and manipulated images. By learning patterns and characteristics associated with forged images, the AI model can make predictions on new images and assist users in determining image authenticity. Therefore, the project contributes to the application of artificial intelligence in solving real-world digital security problems [8], [16].

4. Deep Learning

Deep Learning is an advanced subset of artificial intelligence that uses neural networks to analyze large amounts of data and identify complex patterns. In this project, deep learning models

such as Convolutional Neural Networks (CNNs) and Residual Networks (ResNet) are used for image classification tasks. These models automatically learn hierarchical features from images and can detect subtle differences between authentic and manipulated content. Transfer learning techniques are also applied to improve performance by using pre-trained models and adapting them to the forgery detection task. As a result, the project clearly belongs to the domain of deep learning-based image analysis [17], [18].

5. Web Application Development

In addition to artificial intelligence and computer vision, the project also involves web application development. A web-based interface is developed to allow users to upload images and receive forgery detection results instantly. The backend system integrates the trained deep learning model, while the frontend provides a simple interface for user interaction. By deploying the system as a web application, the project bridges the gap between research and real-world usage, making forgery detection tools accessible through a standard web browser [19].

1.4 Problem Statement

The rapid growth of digital technologies and social media platforms has significantly increased the use of digital images as sources of information, communication, and evidence. Images are widely used in journalism, legal investigations, and online media where authenticity and reliability are important. However, the availability of advanced image editing software and artificial intelligence tools has made it easy to manipulate images in realistic ways, making it difficult to distinguish between genuine and forged images [3], [4].

Various image manipulation techniques such as copy-move forgery, image splicing, object removal, and AI-generated image synthesis allow users to modify images with minimal effort. These manipulations can spread misinformation, damage credibility, and create serious challenges for digital media verification. Detecting manipulated images using human visual inspection alone is often unreliable because forged images may appear highly realistic [5], [6].

Traditional digital image forensic methods analyze metadata, noise patterns, and compression artifacts to identify manipulation. However, these techniques may fail when images undergo common post-processing operations such as resizing, recompression, or format conversion. Deep learning approaches have recently shown promising results in detecting image forgeries by automatically learning complex patterns from large image datasets. Despite their success, many existing models lack interpretability and are difficult to deploy in real-world applications. Most research systems remain confined to academic environments and do not provide user-friendly tools for verifying image authenticity [20], [21].

Therefore, there is a need for an efficient and accessible image forgery detection system

that can accurately identify manipulated images and provide reliable analysis results. This project proposes a deep learning-based image forgery detection system integrated with a web-based application that allows users to upload images and analyze their authenticity through an easy-to-use interface [22].

1.5 Objectives of the Project

The primary aim of this project is to design and develop an intelligent and reliable Image Forgery Detection System capable of identifying manipulated images using deep learning and image forensic techniques. With the increasing availability of image editing tools and artificial intelligence technologies, it has become easier to manipulate digital images in ways that are difficult to detect with the human eye. Therefore, the proposed system focuses on building a robust detection framework that can analyze images, identify potential manipulations, and present clear results to users through a web-based interface. The detailed objectives of the project are explained below [9], [10].

1. Development of a Structured Image Dataset

The primary objective of this project is to develop a dataset containing both authentic and manipulated images for training and evaluating the image forgery detection model. Images are collected from multiple sources and categorized based on different manipulation techniques such as copy-move forgery, image splicing, inpainting, and enhancement-based modifications. The dataset is carefully organized to remove duplicate or corrupted images and ensure proper data quality. After preprocessing, the dataset is divided into training, validation, and testing sets using a 70-15-15 ratio to support effective model training and evaluation. This structured dataset enables the deep learning model to learn various manipulation patterns and improve its ability to detect forged images accurately [12], [13].

2. Development of Deep Learning-Based Forgery Detection Models

Another important objective of this project is to design and train deep learning models capable of accurately detecting forged images. The project utilizes Convolutional Neural Networks (CNNs), which are widely used for image classification and computer vision tasks due to their ability to learn hierarchical visual features. In particular, the system focuses on the ResNet family of architectures, including ResNet-18, ResNet-50, and ResNet-101, which are known for their high performance and efficient feature extraction capabilities [17], [18].

3. Implementation of Image Forensic Analysis Techniques

In addition to deep learning models, another objective of the project is to incorporate image forensic analysis techniques to detect suspicious patterns within digital images. Image forensics focuses on analysing specific characteristics of images such as noise distribution, compression

artifacts, edge inconsistencies, colour distribution, and local pixel variance [1], [2].

4. Implementation of a Hybrid Decision-Making Mechanism

Another objective of this project is to design a hybrid detection mechanism that combines deep learning predictions with results obtained from image forensic analysis techniques. Deep learning models are effective in identifying complex visual patterns, but their decisions are often difficult to interpret. To overcome this limitation, the proposed system integrates neural network predictions with heuristic analysis results to produce a more reliable final decision. This hybrid approach improves detection accuracy and enhances the robustness of the system for real-world image forgery detection [7], [8].

5. Development of a Web-Based User Interface

A major objective of this project is to develop a user-friendly web application that allows users to easily interact with the image forgery detection system. The web-based platform eliminates the need for specialized software and allows users to upload images directly through a browser interface. The web application is developed using Flask for backend processing and HTML, CSS, and JavaScript for the frontend interface. The backend system processes uploaded images, performs necessary preprocessing operations, and sends them to the trained deep learning model for analysis. The results are then displayed to the user in a clear and understandable format [19].

6. Performance Evaluation and Model Testing

Another important objective of this project is to evaluate the performance of the proposed detection system using standard machine learning evaluation metrics. Proper evaluation is essential to ensure that the system produces reliable and accurate results when detecting forged images. The model performance is assessed using metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis. These evaluation metrics help measure how effectively the model distinguishes between authentic and manipulated images [24], [25].

7. Development of a Scalable and Extensible System Architecture

The final objective of this project is to design a scalable and flexible system architecture that can be expanded with additional capabilities in the future. The system is developed using a modular design approach, which allows different components such as the deep learning model, forensic analysis modules, and web interface to be updated independently. This modular architecture enables future improvements such as integrating more advanced deep learning models, expanding the dataset, or adding support for additional multimedia forgery detection tasks such as video forgery detection or deepfake detection [14], [26].

1.6 Tools Required for the Forgery Detection Project

The implementation of the proposed system utilizes several software tools and libraries. Each tool plays an important role in data processing, model development, and performance evaluation [15].

1. Python Programming Language: Python is the primary programming language used for developing the entire deep learning system. It provides a simple and readable syntax, making it easy to implement complex algorithms. Python also supports a wide range of libraries and frameworks specifically designed for data science and machine learning. Its flexibility and strong community support make it highly suitable for building predictive models and handling large datasets efficiently [16].

2. TensorFlow and Keras: TensorFlow is an open-source deep learning framework widely used for building and training neural network models. Keras is a high-level deep learning API built on top of TensorFlow that simplifies the creation of neural networks. In this project, TensorFlow and Keras are used to implement the Convolutional Neural Network (CNN) architecture and support the training process [17].

3. PyTorch: PyTorch is another powerful deep learning framework used for building neural networks. It is widely used in research and development for computer vision tasks. In this project, PyTorch is used to develop and test alternative deep learning architectures for image forgery detection [18].

4. OpenCV and Pillow: OpenCV (Open Source Computer Vision Library) and Pillow are used for image processing tasks such as image loading, resizing, filtering, feature extraction, and preprocessing. They play an important role in preparing the images before feeding them into the deep learning model [7].

5. NumPy and Pandas: NumPy and Pandas are essential libraries used for data preprocessing and numerical operations. NumPy provides support for numerical computations and array operations, while Pandas is used for data manipulation and analysis [24].

6. Matplotlib and Seaborn: Matplotlib and Seaborn are used for data visualization and graphical analysis. These libraries help in representing the results in a clear and understandable manner through graphs and charts. In this project, they are used to plot confusion matrices, performance comparison graphs, and training curves [25], [27].

7. Scikit-learn: Scikit-learn is a machine learning library that provides tools for model evaluation and preprocessing. In this project, it is mainly used for performance evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis [26].

8. Flask Web Framework: Flask is a lightweight Python web framework used to develop the

web application interface for the image forgery detection system. It acts as the backend server that connects the trained deep learning model with the user interface. Through the Flask application, users can upload an image, and the system processes the image through the trained model to determine whether it is authentic or manipulated [19].

9. Visual Studio Code: Visual Studio Code is used as the development environment for writing and executing the project code. It provides features such as syntax highlighting, debugging tools, and an integrated terminal, which make coding more efficient [10].

1.7 Organization Of The Thesis

The thesis is organized into five chapters to present the work in a systematic and structured manner. The first chapter introduces the problem, motivation, objectives, and scope of the study. The second chapter provides a review of existing literature and discusses the background of image forgery detection. The third chapter describes the proposed methodology, including data collection, preprocessing, feature extraction, and implementation of deep learning models. The fourth chapter presents the implementation details and experimental results along with performance evaluation and comparative analysis. The final chapter concludes the study by summarizing the outcomes and highlighting possible directions for future work [2], [3], [22].

CHAPTER – 2

LITERATURE SURVEY

2. Literature survey

2.1 Overview Of Image Forgery Detection

Image Forgery Detection (IFD) is a critical domain within Digital Image Forensics that focuses on identifying manipulated or tampered images based on their visual characteristics and statistical properties [1], [3]. The primary objective is to classify images into categories such as "Authentic" or "Forged" to ensure that digital media can be verified for authenticity in legal, journalistic, and scientific contexts [2], [4].

2.1.1 The Shift from Manual to Automated Detection

Traditionally, image forgery detection was a manual process, where forensic experts examined images for visual inconsistencies using specialized software tools. Modern digital environments now utilize Deep Learning to adopt an automated approach. By analyzing patterns in data such as noise distribution, compression artifacts, and edge inconsistencies, deep learning models can detect forged images with high accuracy, allowing rapid verification of digital content [5], [6].

2.1.2 Key Detection Features

To detect image forgeries accurately, various features are analyzed. In the context of this project, the most significant detection features include:

- **Noise Inconsistencies:** Variations in noise patterns across different regions of an image that may indicate tampering.
- **Compression Artifacts:** Double JPEG compression traces and block artifacts that reveal editing history.
- **Edge Inconsistencies:** Discontinuities in edges and boundaries between spliced regions.
- **Color Distribution:** Unnatural color variations or lighting inconsistencies in manipulated areas.

2.1.3 Significance and Benefits

Detecting image forgeries is vital for reducing misinformation and improving the trustworthiness of digital media [11]. A successful image forgery detection system helps in:

- **Media Verification:** Authenticating images before publication in news and journalism.
- **Legal Evidence Validation:** Ensuring photographic evidence presented in court is authentic.
- **Social Media Content Moderation:** Detecting and flagging manipulated images shared online.
- **Cybersecurity:** Preventing fraud through forged identity documents and insurance claims.

2.2 Reviews Of Existing Works

W. Oettinger (2020) in the book "Learn Computer Forensics: A Beginner's Guide to Searching, Analyzing, and Securing Digital Evidence" provided a comprehensive introduction to computer forensics, covering the fundamental concepts, tools, and techniques used in digital investigations.

This work is essential for understanding the broader context of digital forensics within which image forgery detection operates [1].

N. Parashar and N. Tiwari (2015) in "A Survey of Digital Image Tampering Techniques" provided a comprehensive survey of various digital image tampering techniques including copy-move, splicing, and retouching. Their work serves as a valuable resource for understanding the landscape of image forgery and detection methods [2].

J. Dong, W. Wang, and T. Tan (2013) in "Casia Image Tampering Detection Evaluation Database" introduced the CASIA image tampering detection database, a large-scale dataset for evaluating image forgery detection algorithms. This database has become a standard benchmark for testing detection methods [3].

S. alZahir and R. Hammad (2020) in "Image Forgery Detection Using Image Similarity" presented a forgery detection method based on image similarity analysis. Their method focuses on detecting forged regions by analyzing structural similarities and differences between image patches [4].

D. Afchar et al. (2018) in "Mesonet: A Compact Facial Video Forgery Detection Network" introduced MesoNet, a compact neural network for detecting facial video forgeries and deepfakes. Their model analyzes facial features and movement patterns to identify manipulated video content [5].

D. Patrikar, U. Kosarkar, and A. Chaube (2023) in "Comprehensive Study on Image Forgery Using Deep Learning" provided a comprehensive review of image forgery techniques and detection methods using deep learning. Their study surveys various manipulation techniques and evaluates different deep learning approaches [6].

K. O'Shea and R. Nash (2015) in "An Introduction to Convolutional Neural Networks" provided a comprehensive introduction to CNNs, explaining their architecture and working principles. This foundational work helps understand the mechanisms of deep learning models used in image forensics [7].

I. Goodfellow et al. (2020) in "Generative Adversarial Networks" introduced GANs, a framework where two networks compete to generate realistic synthetic data. This work has significant implications for both creating and detecting synthetic forgeries [8].

G. Koch et al. (2015) in "Siamese Neural Networks for One-Shot Image Recognition" introduced Siamese networks for comparing image similarity. This architecture is particularly useful for

detecting copy-move forgeries by identifying duplicated regions [9].

J. Deng et al. (2009) in "Imagenet: A Large-Scale Hierarchical Image Database" introduced ImageNet, a large-scale database of labeled images that has become essential for training deep learning models. Pre-trained models on ImageNet are widely used for transfer learning in image forgery detection [10].

S. Bozinovski (2020) in "Reminder of the First Paper on Transfer Learning in Neural Networks" discussed the historical development of transfer learning techniques. Transfer learning is crucial for adapting pre-trained models to image forgery detection tasks with limited data [11].

Kaggle Dataset (2023) in "Image Forgery Detection" provided a publicly available dataset containing authentic and forged images for training and evaluating forgery detection models. This dataset is used in the proposed system for model training and testing [12].

K. Simonyan and A. Zisserman (2014) in "Very Deep Convolutional Networks for Large-Scale Image Recognition" introduced the VGG architecture, which demonstrated that increasing network depth improves classification accuracy. VGG models are commonly used as baselines for image forgery detection [13].

S. Xie et al. (2017) in "Aggregated Residual Transformations for Deep Neural Networks" introduced ResNeXt, an extension of ResNet that uses grouped convolutions. Their work improved upon residual networks for better feature extraction [14].

A. Paszke et al. (2019) in "Pytorch: An Imperative Style, High-Performance Deep Learning Library" introduced PyTorch, a deep learning framework that enables flexible model development. PyTorch is widely used for implementing image forgery detection systems [15].

C. Song et al. (2019) in "Image Forgery Detection Based on Motion Blur Estimated Using Convolutional Neural Network" proposed a method for detecting image forgery using motion blur estimation with the help of a CNN. The model learns motion-blur characteristics and identifies abnormal blur regions that may indicate manipulation [16].

J. H. Bappy et al. (2019) in "Hybrid LSTM and Encoder-Decoder Architecture for Detection of Image Forgeries" introduced a hybrid deep learning model that combines LSTM networks with an encoder-decoder architecture. Their system analyzes resampling features to identify manipulations such as image splicing and can locate tampered regions with pixel-level precision [17].

F. Marra et al. (2020) in "A Full-Image Full-Resolution End-to-End-Trainable CNN Framework for Image Forgery Detection" presented a CNN framework that processes full-resolution images without downsampling. Their model detects manipulations by learning noise inconsistencies in forged regions and produces localization maps highlighting tampered areas [18].

L. Zhao et al. (2021) in "Deep Learning-Based Forgery Attack on Document Images" introduced ForgeNet, a model designed to generate realistic forgeries in document images. Their research highlighted the challenges posed by advanced deep learning techniques in digital document security and emphasized the need for stronger detection methods [19].

K. M. Hosny et al. (2022) in "An Efficient CNN Model to Detect Copy-Move Image Forgery" proposed a CNN model that extracts multi-scale features from images and uses a similarity network to identify duplicated regions. Their method can accurately locate copied and pasted regions indicating manipulation [20].

N. Dhiman et al. (2023) in "An Efficient Approach for Image Forgery Detection Using Deep Convolutional Neural Network" presented a deep CNN approach that extracts multi-level features from noise patterns and image structures. Their method detects various types of manipulations such as copy-move and splicing with precise localization [21].

M. H. Mutar et al. (2024) in "Innovative Deep Learning Solutions for Image Forgery Detection" presented deep learning approaches using CNNs and GANs to identify common manipulation techniques including image splicing and copy-move forgery. Their system analyzes noise patterns and combines multiple image features through feature fusion [22].

A. Meepaganithage et al. (2024) in "Image Forgery Detection Using Convolutional Neural Networks" proposed a CNN-based method that analyzes spatial features and noise inconsistencies across image patches. Their model detects copy-move and image splicing manipulations by learning complex patterns from large datasets [23].

C. R. Harris et al. (2020) in "Array Programming with NumPy" discussed NumPy, the fundamental library for numerical computing in Python. NumPy is essential for image preprocessing and data manipulation in forgery detection systems [24].

F. Pedregosa et al. (2011) in "Scikit-learn: Machine Learning in Python" introduced scikit-learn, a machine learning library providing tools for model evaluation. Scikit-learn is used for computing performance metrics such as accuracy, precision, and recall [25].

J. D. Hunter (2007) in "Matplotlib: A 2D Graphics Environment" introduced Matplotlib, a comprehensive visualization library for Python. Matplotlib is used for plotting confusion matrices and training curves in forgery detection systems [26].

M. Waskom et al. (2017) in "Seaborn: v0.8.1" introduced Seaborn, a statistical data visualization library built on Matplotlib. Seaborn is used to create advanced visualizations such as heatmaps and confusion matrices [27].

2.3 Approaches for Image Forgery Detection Using Deep Learning Models

1. Local Feature Extraction Using Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) play a crucial role in detecting image forgeries by extracting detailed local features from digital images. CNN models analyze pixel-level patterns such as texture variations, noise inconsistencies, and edge discontinuities that often appear when an image is manipulated. In forgery detection tasks like copy-move or image splicing, these local inconsistencies help identify tampered regions. The convolution layers automatically learn hierarchical feature representations, enabling the system to detect subtle artifacts left during editing processes. By training CNN models on large image datasets, the system becomes capable of distinguishing between authentic and forged image regions with high accuracy.

2. Similarity Learning Using Siamese Networks

Siamese Networks are deep learning architectures designed to compare two input images and determine their similarity. In image forgery detection, Siamese networks are useful for identifying duplicated regions within an image, which is a common characteristic of copy-move forgery. The Siamese architecture consists of two identical neural network branches that share the same weights and parameters. Each branch processes one input image or image patch and extracts feature representations.

These feature vectors are then compared using a distance metric such as Euclidean distance or cosine similarity. If the similarity between the feature representations is high, the system can identify potential duplicated or manipulated regions. This approach is particularly effective in detecting forgery patterns that involve copying and pasting parts of the same image.

3. Hybrid CNN–Siamese Network Architecture

To improve the accuracy of forgery detection, a hybrid architecture combining CNN and Siamese networks is used. In this approach, the CNN is first used to extract meaningful features from image patches or regions. These extracted features represent the structural and texture characteristics of the image. The Siamese network then compares these feature representations to detect similarities between different regions of the image. By combining feature extraction with

similarity comparison, the system can effectively detect duplicated regions and identify forged areas. This hybrid architecture enhances the model's capability to detect various types of image manipulation with higher reliability.

4. Image Preprocessing Techniques

Before training the deep learning models, image preprocessing is performed to prepare the dataset and improve model performance. The preprocessing stage includes resizing images to a standard input size required by the CNN model. Images are normalized so that pixel values fall within a specific range, which stabilizes the learning process. Noise reduction and contrast enhancement techniques may also be applied to highlight important image features. Additionally, data augmentation methods such as rotation, flipping, cropping, and scaling are used to increase dataset diversity and prevent overfitting during training. These preprocessing steps ensure that the model receives consistent and high-quality input data.

5. Model Training and Evaluation Pipeline

The training process begins with collecting a dataset containing both authentic and forged images. The dataset is divided into training, validation, and testing sets to properly evaluate the model's performance. The CNN and Siamese network architecture is implemented using deep learning frameworks such as TensorFlow or PyTorch. Training is often performed using platforms like Google Colab or Jupyter or VScode Notebook with GPU acceleration to speed up computations. During training, the model learns to extract meaningful features and compare image regions to detect manipulation.

The model's performance is evaluated using metrics such as accuracy precision, recall, and F1-score. These metrics help determine the effectiveness of the proposed system in detecting image forgeries.

6. Web-Based Deployment Using Flask

After training and selecting the best-performing deep learning model, the system is deployed as a web-based application using the Flask framework. The trained CNN-Siamese network model is saved and integrated into the Flask backend for prediction. Users can upload a digital image through a simple HTML-based interface provided by the web application. Once the image is uploaded, it undergoes the required preprocessing and is passed to the trained model for analysis. The system then predicts whether the image is authentic or forged, and the result is displayed instantly on the web page. This web-based deployment makes the image forgery detection system easily accessible and allows users to verify the authenticity of images through a standard web browser.

2.4 System Architecture

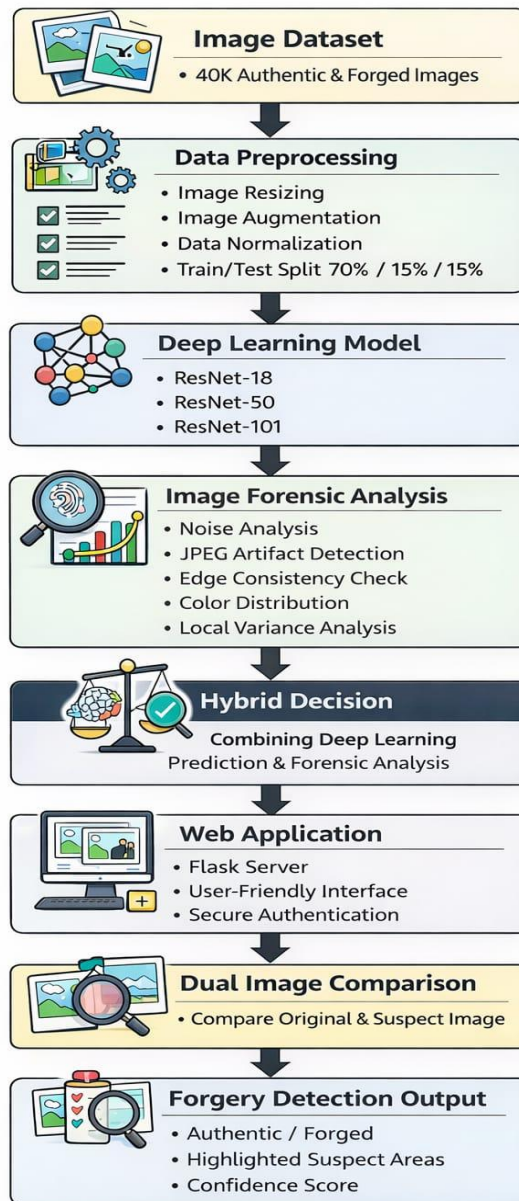


Fig 2.1 : System Architecture of Image forgery Detection

The Image Forgery Detection system is designed to analyze digital images and determine whether an image is authentic or manipulated. The system uses deep learning algorithms, specifically Convolutional Neural Networks (CNN) and Siamese Networks, to learn patterns from images and detect signs of tampering. These models analyze visual features such as textures, edges, noise patterns, and duplicated regions to identify possible image manipulation.

In this project, a dataset containing both genuine and forged images is used to train the deep learning model, including different types of image forgeries such as copy-move forgery, image splicing, and other editing-based manipulations. The trained model is integrated into a Flask-based web application that allows users to upload an image and receive a prediction indicating whether the image is authentic or forged.

The system architecture consists of several stages including dataset collection, image preprocessing, feature extraction using CNN, similarity comparison using Siamese networks, model training, model evaluation, model deployment, and prediction through a web interface. The trained CNN–Siamese model is evaluated using performance metrics such as accuracy, precision, recall, and F1-score, and the best-performing model is deployed in the web application to provide real-time image forgery detection. This system converts raw digital images into meaningful predictions that help identify manipulated images

Dataset Input

The first step in the system architecture is dataset input. In this project, an image dataset containing both authentic and manipulated images is used. The dataset may include several types of image manipulations such as copy-move forgery, image splicing, and other editing techniques. These images act as the primary source of data used to train the deep learning model. The dataset is loaded into the system using Python libraries such as OpenCV, NumPy, and TensorFlow/Keras for further processing and analysis. The dataset serves as the foundation of the entire forgery detection system because the deep learning models learn patterns and inconsistencies from these images to distinguish between genuine and forged images.

Image Preprocessing

After loading the dataset, the next step is image preprocessing. Preprocessing is necessary because raw images may have different sizes, noise levels, or lighting conditions that can affect model performance.

In this project, preprocessing steps include resizing images to a fixed input size, normalization of pixel values, and noise reduction. These steps ensure that all images have a consistent format suitable for deep learning models. Image preprocessing improves the overall quality of the dataset and allows the model to learn more accurately.

CNN Feature Extraction

In this stage, a Convolutional Neural Network (CNN) is used to extract important visual features from the images. CNN models are capable of automatically learning hierarchical patterns from image data. These patterns include edges, textures, noise inconsistencies, and structural differences that may appear when an image has been manipulated.

The convolution layers scan the image using filters to detect important patterns. Pooling layers reduce the dimensionality of the feature maps while preserving important information. This layered structure allows the CNN to learn increasingly complex image features that help distinguish authentic images from manipulated ones. These extracted features are then passed to the Siamese network for further comparison and forgery detection.

Similarity Comparison Using Siamese Network

The Siamese Network is used to detect similarities between different regions of an image. This is particularly useful for identifying copy-move forgeries, where a portion of the image is copied and pasted into another region.

The Siamese network consists of two identical neural network branches that share the same parameters. Each branch processes an image patch and extracts its feature representation. The system then compares the two feature vectors using a similarity metric such as Euclidean distance. If the similarity between two image regions is high, it may indicate that the region has been duplicated, suggesting possible forgery. This approach helps the system detect manipulated areas even when the copied region has been slightly modified.

Model Training

In the model training stage, the hybrid CNN–Siamese network is trained using the prepared image dataset. The CNN extracts visual features from images, while the Siamese network learns to compare image regions and detect duplicated patterns. During training, the dataset is divided into training and testing sets. The model learns the differences between authentic and manipulated images by adjusting its internal parameters to minimize prediction errors. The training process helps the system develop the ability to automatically detect forgery patterns within digital images.

Model Evaluation

After training the model, its performance is evaluated using several evaluation metrics. These metrics include Accuracy, Precision, Recall, and F1-Score. Accuracy measures how many predictions made by the model are correct. Precision measures how many images predicted as forged are actually manipulated. Recall measures how effectively the model detects forged images

among all manipulated images. The F1-score provides a balanced evaluation of precision and recall. Additional evaluation methods such as confusion matrix analysis can also be used to visualize model performance. The model with the best overall performance is selected as the final forgery detection model.

Model Deployment

Once the best-performing model is selected, it is saved using Python serialization methods such as HDF5 (.h5) or Pickle. The saved model is then integrated into a Flask-based web application. Flask acts as the backend framework that connects the trained Hybrid CNN-Siamese network model with the user interface. Users can upload an image through the web page, and the system processes the image using the trained model. The model analyzes the image features and predicts whether the uploaded image is authentic or forged. The prediction result is displayed instantly on the web page, making the system a practical tool for image forgery detection.

Prediction Through Web Interface

The final stage of the system architecture is prediction through a web interface. A user-friendly web page is created using HTML and CSS, where users can upload an image for verification. Once the user uploads an image, the file is sent to the Flask server. The server performs the necessary preprocessing steps and passes the processed image to the trained CNN-Siamese model. The model analyzes the image and predicts whether the image is authentic or forged. The prediction result, along with confidence information, is displayed on the result page. This interactive web interface makes the system easy to use for digital forensic investigators, journalists, researchers, and general users who want to verify the digital images.

2.5 Software Used

2.4.1 Python Programming Language

Python is the primary programming language used in this project and is widely applied in machine learning, deep learning, and data science due to its simplicity, readability, and powerful libraries. Python provides several libraries that simplify machine learning and image processing tasks. In this project, Python is used to load and manage the image dataset, preprocess images, train deep learning models, evaluate model performance, and deploy the trained model in a web-based application. Its flexibility, strong community support, and extensive library ecosystem make Python an ideal choice for developing intelligent systems such as image forgery detection, where deep learning techniques are used to identify manipulated or fake images.

Why Use Python?

Python is one of the most widely used programming languages for machine learning, data science, and web application development. It is known for its simple syntax, readability, and powerful ecosystem of libraries. In this project, Python is used to develop the deep learning model and integrate it with a web application for detecting forged images. Python provides strong support for machine learning and image processing through libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and OpenCV. These libraries help perform tasks such as image preprocessing, feature extraction, model training, testing, and evaluation efficiently. Python also offers platform independence and strong community support, allowing applications to run on different operating systems and enabling developers to easily find resources and solutions. Because of these advantages, Python is an ideal choice for developing intelligent systems such as the Image Forgery Detection system.

2.4.2 Flask Web Framework

Flask is a lightweight web framework used to build web applications in Python. In this project, Flask is used to connect the trained deep learning model with a user interface. The Flask framework handles user requests, processes uploaded image data, and sends prediction results back to the web page. It acts as a bridge between the frontend (HTML interface) and the backend machine learning model. Flask allowing users to upload images and receive results indicating whether the image is authentic or manipulated.

Why Use Flask?

Flask is a lightweight and flexible web framework written in Python that is widely used for building web applications and APIs. In this project, Flask is used to create the web interface that connects the trained deep learning model with the user. It acts as a bridge between the frontend interface and the backend machine learning model, allowing users to upload images and receive detection results through a web browser. Flask is easy to configure and integrates well with Python libraries, making it suitable for deploying machine learning models as web applications. It handles user requests, processes uploaded images, sends them to the trained model for analysis, and displays the prediction results indicating whether the image is authentic or manipulated.

2.4.3 Setting up Visual Studio

Visual Studio Code is a lightweight and powerful source code editor used for developing software applications. It supports multiple programming languages such as Python, HTML, CSS, and JavaScript. In this project, Visual Studio Code is used to write Python scripts for implementing the deep learning model and to develop the web application using the Flask framework. It provides useful features such as syntax highlighting, code suggestions, debugging

support, and extensions that help developers write and manage code efficiently. Its speed, flexibility, and support for various development tools make it suitable for developing projects like the Image Forgery Detection system.

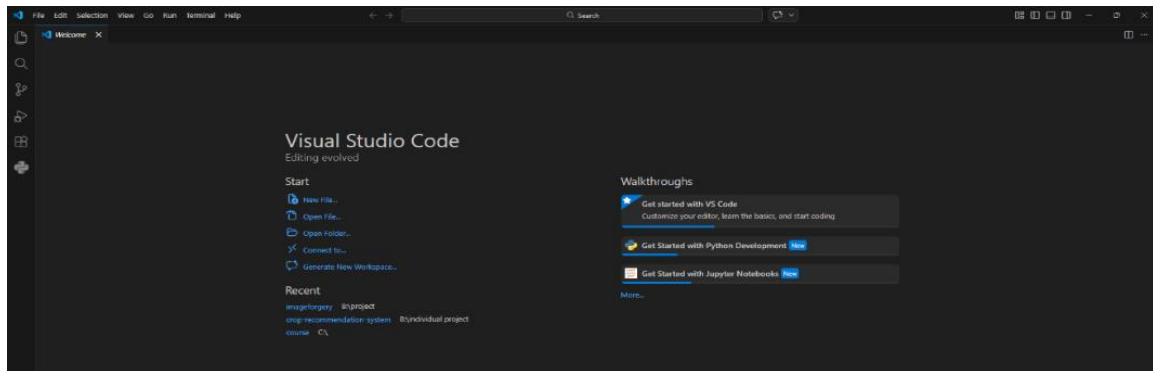


Fig 2.2 : Opening Visual Studio Code

The interface contains a large editing area where the code is written, along with menu options such as File, Edit, View, Run, and Terminal for managing project files.

Creating a Project Folder in Visual Studio Code

Organizing project files properly is an important step when developing any software application. In the Image Forgery Detection project, a dedicated project folder is created to store all the required files such as Python scripts, HTML templates, static resources, trained models, and image datasets used for training and testing the system.

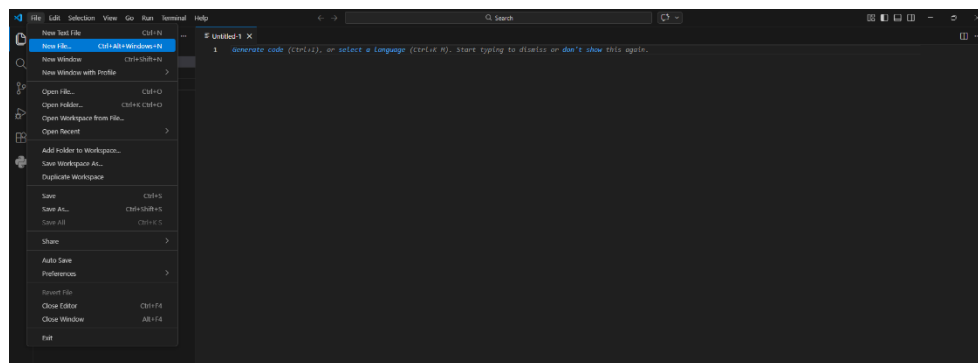


Fig 2.3 : Creating a Project Folder.

Inside this folder, different files and subfolders are created to organize the application structure. For example:

app.py :- Contains the Flask web application code.

train_model.py :- Contains the deep learning model training code.

templates folder :- Stores HTML files used for the web interface.

static folder :- Stores CSS files, images, and other resources.

dataset folder :- Contains authentic and forged images used for training and testing the model.

Organizing the files in this structured way helps developers easily manage the project and maintain a clear workflow during development.

Writing Python Code in Visual Studio Code

After setting up the project folder, the next step is writing the Python code required for the application. Visual Studio Code provides a clean and efficient coding environment where developers can write, edit, and save Python programs easily.

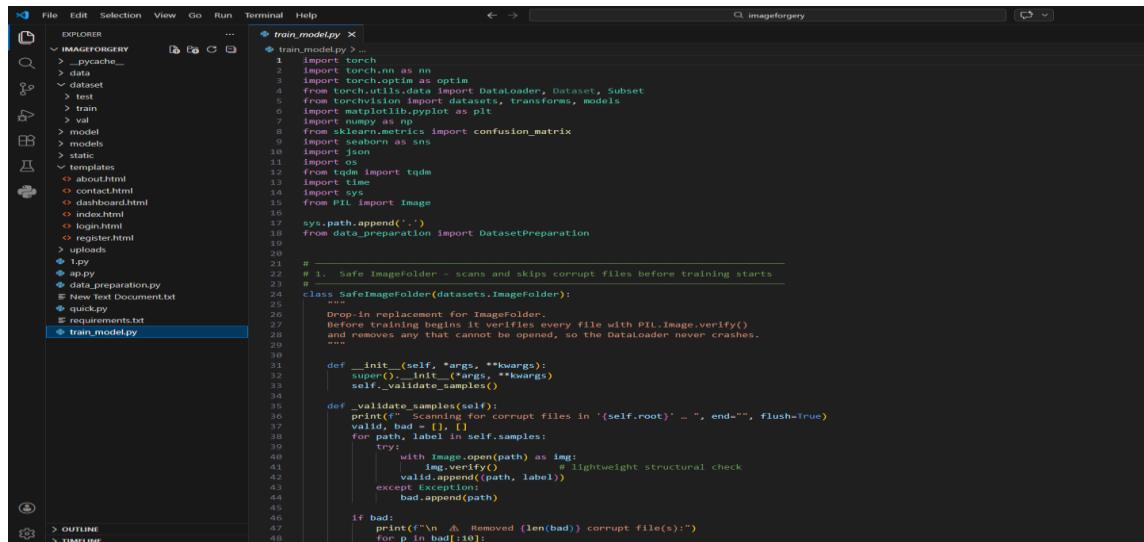


Fig 2.4 : Writing Python Code in Visual Studio Code

Figure 2.4 shows the Visual Studio Code editor where the Python code for the project is written. The developer creates a new file and saves it with the “.py” extension, which indicates that it is a Python program.

In the Image Forgery Detection project, the Python scripts contain the deep learning model implementation and the Flask framework code that connects the trained model with the web interface. Visual Studio Code highlights different parts of the code using colours through syntax highlighting, which helps developers understand the structure of the program more easily and improves coding efficiency.

Running the Project Using Command Prompt (CMD)

The Command Prompt (CMD) is used to execute Python scripts and run the Flask-based Image Forgery Detection system. It serves as an interface between the user and the operating system, allowing developers to navigate folders and run programs.

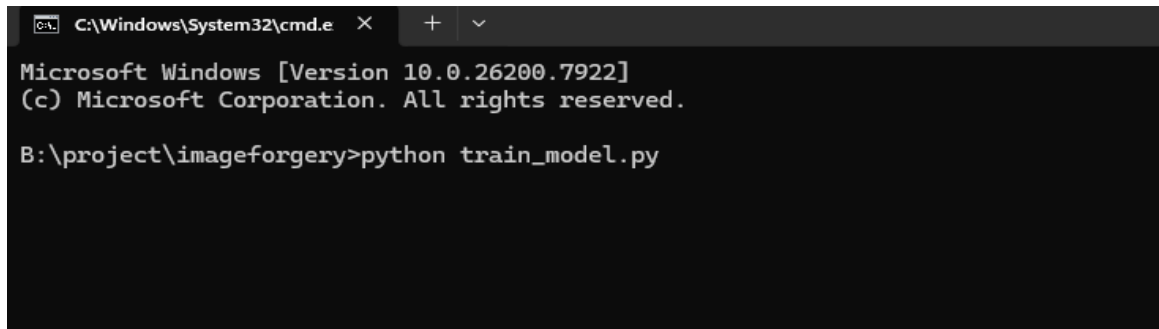
A screenshot of a Windows Command Prompt window. The title bar shows 'C:\Windows\System32\cmd.e' with a close button. The window content displays the following text: 'Microsoft Windows [Version 10.0.26200.7922] (c) Microsoft Corporation. All rights reserved. B:\project\imageforgery>python train_model.py'. The background is black, and the text is white.

Fig 2.5 : Running the Python Program in Command Prompt.

The image 2.5 illustrates the process of running the Image Forgery Detection project using the Command Prompt. First, the user navigates to the project folder using the `cd` (change directory) command. After entering the project directory, the Python file is executed using the command:

```
python app.py
```

Once the command is executed, the Flask server starts running locally on the system. The terminal displays a message showing the local server address, usually `http://127.0.0.1:5000/`.

The user can open this address in a web browser to access the web application. The web interface allows users to upload an image, and the system analyzes the image using the trained model to determine whether it is authentic or forged.

2.4.4 Python Libraries Used in the Project

Python provides a wide range of powerful libraries that help developers perform image processing, deep learning, data analysis, and web application development efficiently. In this project, several Python libraries are used to develop the Image Forgery Detection system. These libraries support tasks such as image preprocessing, model training, data analysis, visualization, and integration of the trained model with a web interface.

The main libraries used in this project include Flask, NumPy, Pandas, TensorFlow, Keras, OpenCV, Pillow, Matplotlib, and PyTorch. Each of these libraries plays an important role in different stages of the project



Fig 2.6 : Python Imaging Libraries

1. Flask

Flask is a lightweight Python web framework used to develop the web application interface for the image forgery detection system. It acts as the backend server that connects the trained deep learning model with the user interface. Through the Flask application, users can upload an image, and the system processes the image through the trained model to determine whether it is authentic or manipulated. Flask also manages routing, request handling, and communication between the frontend and backend.

2. Flask-SQLAlchemy

Flask-SQLAlchemy is used for integrating database functionality within the Flask application. It provides an Object Relational Mapping (ORM) system that simplifies database operations. In this project, it can be used to store user information, uploaded image details, prediction results, and system logs for future reference.

3. Flask-Login

Flask-Login is used to manage user authentication and session management. It ensures that only registered users can access the system and upload images for forgery detection. The library maintains secure login sessions and manages user access control throughout the application.

4. Flask-Bcrypt

Flask-Bcrypt is used to enhance security by encrypting user passwords before storing them in the database. Password hashing ensures that even if the database is compromised, the original passwords remain protected.

5. Werkzeug

Werkzeug is a powerful Python library that provides utilities for web application development. It supports Flask by handling HTTP requests, URL routing, file uploads, and secure filename processing when users upload images to the system.

6. NumPy

NumPy is a fundamental library for numerical computing in Python. It provides support for large multidimensional arrays and matrices that are commonly used in image processing and deep learning operations. NumPy enables efficient mathematical computations required during image preprocessing and model training.

7. Pandas

Pandas is used for data manipulation and analysis. In this project, it can be used to organize image metadata, preprocessing logs, and experimental results. It also helps in managing datasets and evaluating model performance during experimentation.

8. TensorFlow

TensorFlow is an open-source deep learning framework widely used for building and training neural network models. In this project, TensorFlow is used to implement the Convolutional Neural Network (CNN) architecture and support the training process of the hybrid CNN–Siamese model for detecting image forgery patterns.

9. Keras

Keras is a high-level deep learning API built on top of TensorFlow that simplifies the creation of neural networks. It provides user-friendly modules for defining layers, activation functions, optimizers, and loss functions. In this project, Keras is used to design and train the CNN architecture and manage the training process efficiently.

10. OpenCV

OpenCV (Open Source Computer Vision Library) is used for image processing tasks such as image loading, resizing, filtering, feature extraction, and preprocessing. It plays an important role in preparing the images before feeding them into the deep learning model. OpenCV can also help in detecting inconsistencies or artifacts that may indicate image manipulation.

11. Pillow

Pillow is a Python imaging library used for opening, manipulating, and saving image files in different formats. It is useful for handling uploaded images within the web application and performing basic image operations such as resizing and format conversion.

12. Matplotlib

Matplotlib is a data visualization library used to generate graphs and charts during model

evaluation. It is used to visualize training accuracy, loss curves, and other performance metrics that help in analyzing the effectiveness of the deep learning model.

13. Seaborn

Seaborn is a statistical data visualization library built on top of Matplotlib. It is used to create advanced visualizations such as confusion matrices, correlation plots, and heatmaps that help in evaluating the performance of the forgery detection model.

14. Scikit-learn

Scikit-learn is a machine learning library that provides tools for model evaluation and preprocessing. In this project, it is mainly used for performance evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis.

15. Joblib

Joblib is used for saving and loading trained machine learning models efficiently. After training the CNN or hybrid model, it can be serialized using Joblib so that the model can be directly loaded into the Flask application without retraining.

16. Scikit-Image

Scikit-Image is a Python library used for advanced image processing and feature extraction. It provides functions for filtering, segmentation, and image transformation that can help detect subtle differences between authentic and forged images.

17. PyTorch

PyTorch is another powerful deep learning framework used for building neural networks. It is widely used in research and development for computer vision tasks. In some experiments, PyTorch can be used to develop or test alternative deep learning architectures for image forgery detection.

CHAPTER-3

PROPOSEDSYSTEM

3. Proposed Methodology

3.1 Introduction to the Methodology

The proposed system is designed to detect manipulated or forged images using deep learning techniques combined with heuristic forensic analysis. The system integrates image preprocessing, deep learning classification, forensic quality analysis, and a web-based interface to provide a complete image forgery detection solution. The process begins when a user uploads an image through a web interface developed using HTML and CSS. The uploaded image is received by the backend application built using the Flask framework, where preprocessing operations such as resizing and normalization are performed to prepare the image for analysis. The preprocessed image is then passed to a trained deep learning model based on the ResNet architecture, which analyzes visual patterns and predicts whether the image is authentic or forged. In addition to deep learning classification, the system performs heuristic forensic analysis including noise analysis, compression artifact detection, and color distribution analysis. The results from these analyses are combined using a hybrid decision mechanism to generate the final classification along with a confidence score. The prediction result and analysis details are then displayed to the user through the web application.

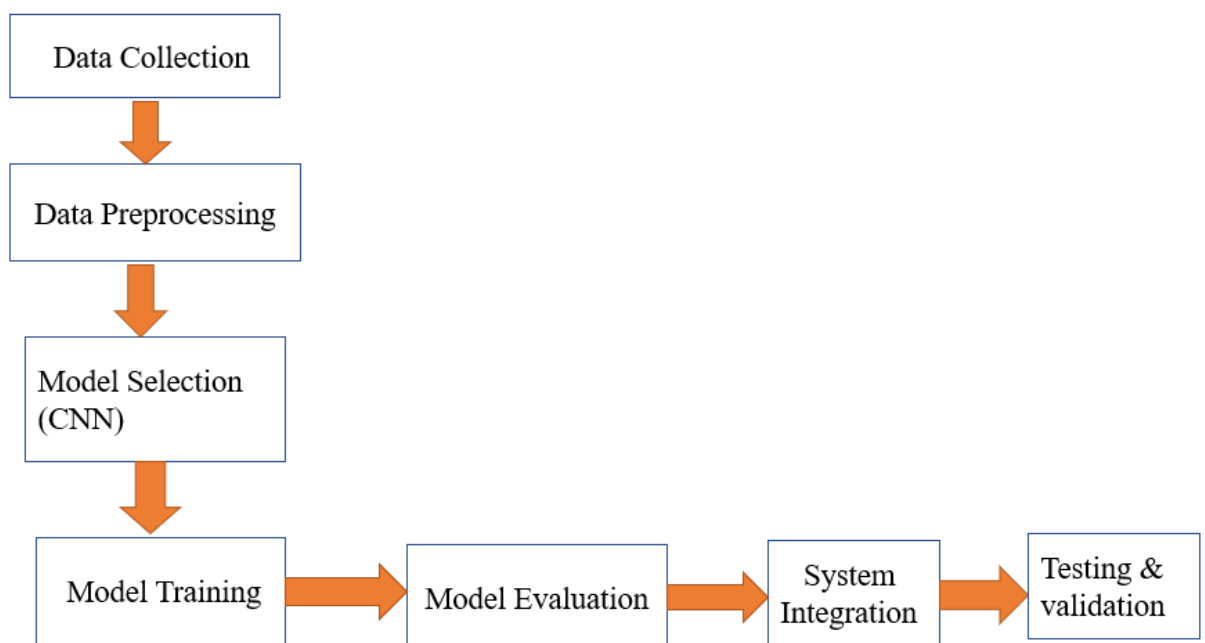


Fig 3.1 : Proposed Methodology

3.2 User

The process begins with the user interacting with the system through the web interface. The user uploads an image that needs to be analyzed for possible manipulation or forgery. This image acts as the primary input to the image forgery detection system. Once the image is uploaded, it is transferred to the backend server where further processing and analysis are performed.

3.3 Index Page (index.html)

The index.html page acts as the front-end interface of the system. It provides a simple and user-friendly interface where users can upload an image file for forgery detection.

The page is developed using HTML and CSS to ensure ease of use and accessibility. It contains a file upload option and a button that allows users to submit the image for analysis. When the user clicks the upload or detect button, the image is sent to the Flask backend server through an HTTP POST request.

3.4 Flask Application (app.py)

The Flask application functions as the backend controller of the system. It receives the uploaded image from the HTML form and processes it for analysis. The backend application loads the trained deep learning model and necessary preprocessing tools. It also performs image validation and converts the uploaded image into the appropriate format required for model prediction. The Flask server manages communication between the user interface and the detection system.

3.5 Image Preprocessing

Before sending the image to the deep learning model, preprocessing operations are applied to ensure compatibility with the model architecture.

The uploaded image is resized to 224×224 pixels, which is the required input size for the ResNet model. Pixel values are then normalized to match the distribution used during model training. This preprocessing step ensures that the model receives consistent input data and improves prediction accuracy.

3.6 Deep Learning Model

The preprocessed image is passed to the trained deep learning model for classification. The model used in this project is based on the ResNet architecture, which is known for its strong performance in image recognition tasks.

Different variants such as ResNet-18, ResNet-50, and ResNet-101 were evaluated during the training phase. These models were trained using transfer learning with pre-trained weights from the ImageNet dataset. The final fully connected layer was modified to classify images into two categories: Authentic and Forged.

The model analyzes patterns, textures, and structural inconsistencies in the image to determine whether manipulation has occurred.

3.7 Heuristic Image Analysis

In addition to deep learning classification, the system performs heuristic forensic analysis to detect signs of manipulation.

Several image quality metrics are calculated, including noise level, compression blockiness, edge consistency, and color distribution. Noise analysis measures abnormal variations in pixel intensity, while compression analysis detects artifacts that occur due to repeated image editing and saving.

Edge detection techniques are used to examine structural inconsistencies, and color variance analysis identifies unnatural color distributions that may indicate image tampering. These metrics help identify potential signs of forgery that may not always be detected by the neural network alone.

3.8 Hybrid Decision Fusion

The hybrid decision mechanism combines the results from the deep learning model and heuristic analysis to produce a final prediction. If the heuristic analysis strongly indicates manipulation, the system classifies the image as forged even if the neural network prediction is uncertain. When both methods agree, the system produces a highly confident prediction. If heuristic evidence is weak, the final decision mainly depends on the neural network output. This hybrid approach improves the reliability of the system and reduces false predictions.

3.9 Result Page

The final detection result is displayed on the result page of the web application. The result page shows whether the uploaded image is Authentic or Forged along with a confidence score. Additional analysis information such as noise level, compression artifacts, and other forensic indicators may also be displayed to provide better understanding of the detection process. The result page also allows the user to return to the upload page and analyze another image if required.

CHAPTER-4

RESULTSAND DISCUSSION

4. Results and Discussion

4.1 Introduction to Results and Discussion

The Results and Discussion section presents the performance evaluation and effectiveness of the proposed Image Forgery Detection System developed using deep learning techniques and heuristic forensic analysis. After implementing the proposed methodology, several experiments were conducted to evaluate the ability of the system to accurately classify images as authentic or forged.

The dataset used for training and evaluation consists of authentic and manipulated images representing various types of image forgeries such as copy-move manipulation, splicing, and other editing operations. These images were organized into training, validation, and testing subsets to ensure reliable model evaluation.

To detect image manipulation effectively, deep learning models based on the residual network architecture were used. These models were trained using transfer learning techniques and evaluated using standard classification performance metrics such as accuracy, precision, recall, specificity, and F1-score.

The experimental results demonstrate that the developed system can successfully analyze image patterns and identify manipulation artifacts with high reliability. In addition to deep learning classification, heuristic forensic analysis techniques were incorporated to further strengthen the detection capability of the system.

This section also discusses the performance comparison of different deep learning models, evaluation metrics, robustness of the system, and practical applicability of the developed forgery detection platform.

4.2 Evaluation Metrics

Evaluation metrics are used to measure the effectiveness, reliability, and accuracy of the proposed Image Forgery Detection System. These metrics help analyze how accurately the deep learning model can distinguish between authentic and manipulated images. Since the dataset contains both real and forged images, appropriate statistical measures are required to evaluate the model's detection capability.

Several commonly used classification metrics are applied in this project, including accuracy, precision, recall, specificity, and F1-score. These metrics provide a comprehensive understanding of the model's ability to correctly detect forged images while maintaining reliable performance.

Accuracy measures the overall correctness of the model by calculating the proportion of correctly classified images out of the total number of images evaluated. Precision measures how many of the images predicted as forged are actually manipulated, indicating the reliability of the

model when identifying forged images. Recall, also known as sensitivity or true positive rate, evaluates how effectively the system detects actual forged images in the dataset. A high recall value indicates that the system successfully identifies most manipulated images.

The F1-score represents the harmonic mean of precision and recall, providing a balanced evaluation of the model's performance by considering both false positives and false negatives. Specificity measures the ability of the system to correctly identify authentic images, ensuring that genuine images are not incorrectly classified as forged.

To further analyze model performance, a confusion matrix is used. The confusion matrix compares actual class labels with predicted labels and includes four components: true positives, true negatives, false positives, and false negatives. These values help calculate the evaluation metrics and provide deeper insight into the classification performance of the model.

By analyzing these evaluation metrics and the confusion matrix, the effectiveness of the proposed Image Forgery Detection System can be validated, demonstrating its capability to accurately detect manipulated images while maintaining reliable classification performance.

Confusion Matrix-RESNET18:

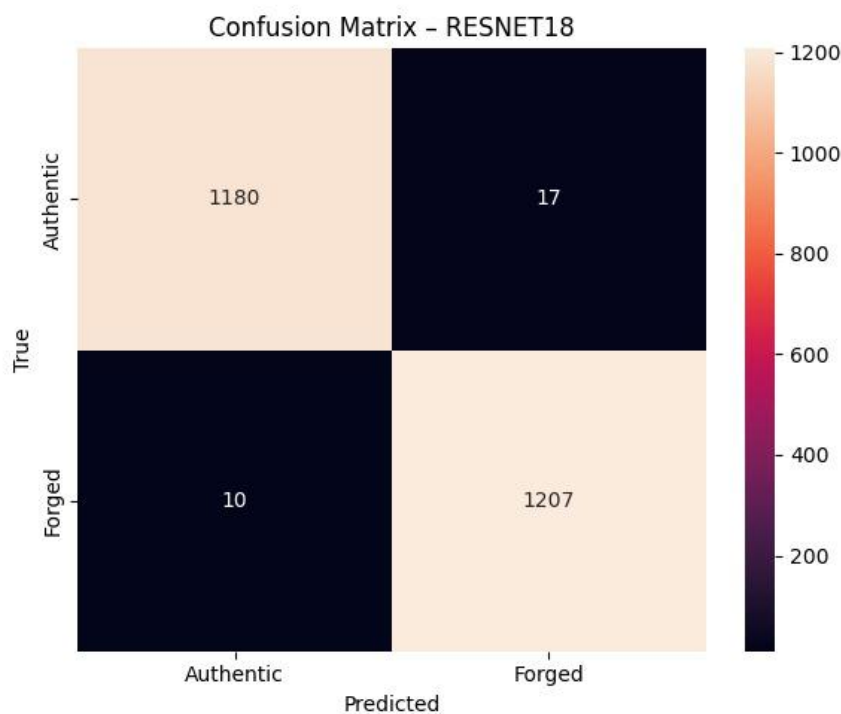


Fig 4.1 : Confusion Matrix of ResNet18

1. Model Performance Metrics

The performance of the proposed image forgery detection system has been evaluated using several standard classification metrics. These metrics help determine how accurately the deep learning model predicts whether an image is authentic or forged based on the visual features present in the dataset.

The confusion matrix is an important tool used to visualize the performance of the classification model. It provides a clear comparison between the actual class labels and the predicted class labels generated by the model. The confusion matrix consists of four main components that help evaluate the effectiveness of the detection system.

- **True Positives (TP = 1207):** These represent the number of forged images that were correctly predicted by the model as forged.
- **True Negatives (TN = 1180):** These represent the number of authentic images that were correctly classified by the model as authentic.
- **False Positives (FP = 17):** These are authentic images that were incorrectly predicted by the model as forged.
- **False Negatives (FN = 10):** These represent forged images that were incorrectly predicted by the model as authentic.

Based on these values from the confusion matrix, several important performance metrics such as accuracy, precision, recall, and F1-score are calculated. These metrics provide a detailed understanding of the model's ability to correctly identify forged images while minimizing incorrect predictions. The evaluation results indicate that the proposed model performs effectively in classifying images and can assist in identifying manipulated images in digital media.

Accuracy

Accuracy is the ratio of correct predictions to the total number of predictions. It represents the overall effectiveness of the model in correctly identifying authentic and forged images from the dataset.

Formula:

$$\begin{aligned}\text{Accuracy} &= (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \\ &= (1207 + 1180) / (1207 + 1180 + 17 + 10) \\ &= 2387 / 2414 \\ &\approx 96.88\%\end{aligned}$$

This accuracy value demonstrates that the model is capable of correctly predicting most of the image classes present in the dataset. It indicates that the deep learning model has successfully learned patterns related to image manipulation and can classify images with a high level of

reliability.

Precision

Precision is the ratio of true positive predictions to the total predicted positives. It tells us how many of the images predicted as forged were actually forged. Precision is especially important in image forgery detection because incorrect predictions may falsely label authentic images as manipulated.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Precision} = 1207 / (1207 + 17)$$

$$\text{Average Precision} \approx 96.61\%$$

High precision means that when the model predicts an image as forged, there is a high probability that the prediction is correct. This improves the reliability of the forgery detection system.

Recall (Sensitivity)

Recall, also known as sensitivity or true positive rate, measures the ability of the model to correctly identify all actual forged images present in the dataset. It is an important metric in forgery detection systems where missing a manipulated image could lead to serious consequences.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{Recall} = 1207 / (1207 + 10)$$

$$\text{Average Recall} \approx 98.18\%$$

A good recall value indicates that the model successfully detects most of the forged images, ensuring that manipulated images are not overlooked during the detection process.

F1 Score

The F1 Score is the harmonic mean of precision and recall. It provides a balanced measure that considers both false positives and false negatives. The F1 Score is especially useful when evaluating classification models where both precision and recall are equally important

Formula:

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$\text{F1 Score} \approx 98.89\%$$

A balanced F1 Score indicates that the model maintains a good balance between precision and recall. This shows that the system performs consistently when detecting authentic and forged images and can be considered a reliable model for image forgery detection.

2. Model Robustness and Interpretation

The evaluation metrics obtained from the confusion matrix demonstrate the effectiveness and robustness of the proposed Image Forgery Detection System. The dataset used in this project contains a large number of images categorized as authentic and forged. The low number of misclassifications indicates that the model is capable of identifying meaningful visual patterns and accurately distinguishing between genuine and manipulated images. The robustness of the system is mainly achieved through the use of deep learning models and proper image preprocessing techniques. Preprocessing methods such as image resizing, normalization, and data augmentation help standardize the input images, enabling the model to learn important features more effectively. Additionally, the design of the proposed system contributes significantly to its performance. Deep learning models such as ResNet-18 and ResNet-50 capture complex visual patterns related to image manipulation, while transfer learning improves detection capability by utilizing knowledge from large pre-trained datasets. These techniques, along with effective preprocessing methods, allow the system to reliably detect forged images while maintaining consistent and accurate performance.

3. Scalability and Real-World Application

With the achieved performance and reliability, the proposed Image Forgery Detection System can be applied in several real-world scenarios where verifying image authenticity is important. The system is implemented using a web-based interface developed with the Flask framework, allowing users to upload images and receive forgery detection results instantly.

Potential applications of this system include:

- **Digital Forensics Investigations** for detecting manipulated images used in cybercrime or legal cases.
- **Media and Journalism Verification** to identify fake or edited images before publishing news content.
- **Social Media Content Moderation** to detect misleading or manipulated images shared online.

Furthermore, the architecture of the system is scalable and can be extended to work with larger and more diverse image datasets. By training the model with additional images and different types of manipulation techniques, the detection accuracy can be further improved. This scalability makes the system suitable for integration into real-world digital media verification and forensic analysis environments.

4. Training and Validation Performance

To evaluate how effectively the deep learning models learned from the dataset, training and validation processes were conducted during the model development stage. The image dataset was

divided into training and testing subsets, where the training data was used to train the model and the testing data was used to evaluate its performance on unseen images.

- The training process allows the model to learn important visual features and patterns associated with authentic and forged images.
- The testing phase evaluates how well the model generalizes to new images that were not included in the training dataset.
- Evaluation metrics such as accuracy, precision, recall, and F1-score are used to measure whether the model performs consistently across both image classes.

A well-performing model should maintain high detection accuracy while minimizing classification errors. The experimental results indicate that the selected deep learning models successfully capture manipulation patterns in images and produce reliable predictions for image forgery detection.

5. Real-Time Performance Evaluation

Real-time performance evaluation measures how efficiently the developed system operates when deployed in practical environments. In the Image Forgery Detection System, this involves testing how quickly the system processes uploaded images and generates detection results through the web interface. One of the key aspects of real-time performance is response time, which refers to the time taken by the system to analyze an image and produce the classification result. The developed system integrates the trained deep learning model with a web application built using the Flask framework, allowing users to upload images and receive detection results quickly.

Tests conducted during system implementation show that the prediction process occurs within a short time, making the system suitable for real-time image analysis. This fast response improves user experience by enabling quick verification of image authenticity.

Another important factor is **system stability**, which refers to the ability of the application to handle multiple image uploads without performance degradation. During testing, the system successfully processed multiple images while maintaining consistent detection accuracy and response time.

Overall, the real-time evaluation confirms that the proposed Image Forgery Detection System is accurate, efficient, and reliable for practical applications. The integration of deep learning models with a lightweight web framework enables the system to provide fast and dependable forgery detection results.

6. Comparison with Existing Methods

The proposed Image Forgery Detection System demonstrates improved effectiveness and reliability when compared with several traditional approaches used for detecting manipulated

images. By integrating deep learning models with heuristic forensic analysis techniques, the system provides accurate and efficient detection of forged images in digital environments.

Manual and Expert-Based Verification:

Earlier image verification methods relied mainly on manual inspection by experts such as journalists, investigators, or photo editors. These methods depend on visual observation to detect inconsistencies in lighting, shadows, or object placement. However, human judgment can be subjective and unreliable when dealing with sophisticated manipulations created using modern editing tools. In addition, manual verification is time-consuming and not suitable for analyzing large numbers of images shared on digital platforms.

Traditional Image Forensics Tools:

Traditional digital forensics tools use signal processing and statistical techniques to identify image manipulation. Tools such as Forensically analyse compression artifacts, metadata information, and image noise patterns to detect possible forgery. Although these methods can detect certain types of manipulation, they often require expert interpretation and may produce inaccurate results when images are heavily compressed or modified using advanced editing software

Early Machine Learning Approaches:

With the advancement of machine learning, algorithms such as Support Vector Machine (SVM), Random Forest, and other classifiers were applied to detect forged images. These methods improved detection performance by learning patterns from image datasets.

However, they relied on handcrafted feature extraction, which limited their ability to generalize across different manipulation techniques and diverse datasets.

Proposed Deep Learning Approach:

In this project, deep learning architectures such as ResNet-18, ResNet-50, and ResNet-101 were implemented and evaluated using performance metrics including accuracy, precision, recall, and F1-score. Among these models, ResNet-101 achieved the highest performance, reaching an accuracy of approximately **93.46%** on the test dataset.

In addition, the system incorporates heuristic forensic analysis techniques such as noise analysis, compression artifact detection, edge consistency analysis, and colour distribution evaluation to strengthen detection reliability.

Compared with existing approaches, the proposed system provides several advantages. It achieves higher detection accuracy, supports detection of multiple types of image manipulation, and improves reliability by combining deep learning with forensic analysis techniques. Furthermore, integration with a web application built using the Flask framework allows users to

upload images and obtain detection results quickly, making the system suitable for real-world digital media verification applications.

4.3 Model Performance Analysis

The performance of the proposed Image Forgery Detection System was evaluated using different deep learning models. In order to identify the most suitable architecture for detecting forged images, multiple models were trained and tested on the image dataset. Each model learns visual patterns and features from authentic and manipulated images to classify whether an image is genuine or forged.

The deep learning models used in this project include ResNet-18, ResNet-50, and ResNet-101 architectures. These models were selected because deep residual networks are highly effective in image classification tasks and can capture complex visual features associated with image manipulation.

To measure and compare the performance of these models, different evaluation techniques were applied. The results were analyzed using confusion matrices, accuracy, precision, recall, and F1-score metrics. These evaluation methods help determine which model performs better in terms of detection accuracy and reliability.

Among the evaluated models, ResNet-101 achieved the highest performance, showing better capability in identifying forged images compared to the other architectures. The results demonstrate that deeper neural network architectures can effectively capture complex forgery patterns and improve the accuracy of image manipulation detection systems.

1. Model Comparison using F1 Score

The figure 4.1 shows the comparison of different deep learning models based on their F1-score values

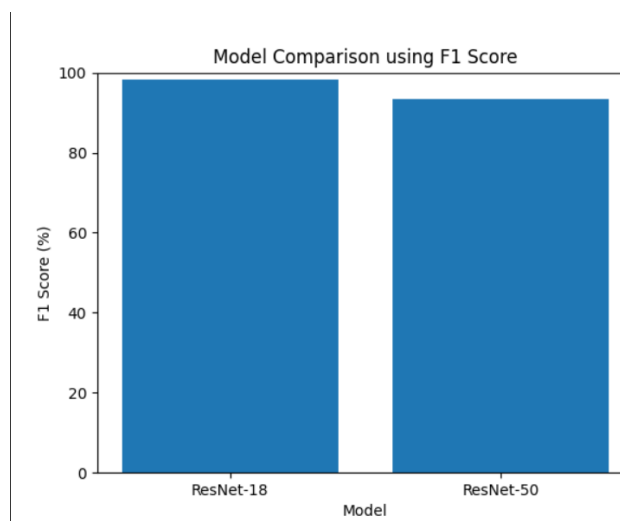


Fig 4.2 : Model Comparison using F1 Score.

The F1 score is an important performance metric that combines both precision and recall into a single value. It is particularly useful when evaluating classification models where both false positives and false negatives need to be minimized. In image forgery detection systems, this metric becomes very important because incorrect predictions may lead to authentic images being flagged as manipulated or, more critically, forged images being accepted as real.

From the comparison, it can be observed that the deep learning models achieved varying F1 scores, which indicates differences in their classification performance. These values suggest that some models are more capable of correctly classifying images while maintaining a balance between precision and recall. Among all the models used in the system, ResNet-18 achieved the highest F1 score of 98.3% .

This indicates that ResNet-18 is highly capable of capturing the visual patterns related to image manipulation in the dataset. Its strong performance demonstrates that a well-tuned architecture, even with fewer layers, can be extremely effective for this task. The ResNet-50 model achieved an F1 score of 93.4% . While this is still a strong performance, it is lower than that of ResNet-18.

This suggests that for this specific dataset, the increased depth and complexity of ResNet-50 may lead to overfitting or difficulty in generalizing, preventing it from matching the balanced performance of its smaller counterpart.

Overall, the F1 score comparison clearly indicates that ResNet-18 provides the best balance between precision and recall for image forgery detection in this study, making it the most suitable model among those evaluated.

2. Training and Validation Performance Analysis

Analyzing the training and validation performance over successive epochs is an important technique used to evaluate deep learning models. These curves help The validation loss started at 0.13 in epoch 1 and remained relatively stable between 0.13 and 0.17 for most of the training process. However, from epoch 8 onwards, a slight increase in validation loss can be observed, reaching 0.21 in epoch 11. This divergence between training loss (decreasing) and validation loss (increasing) in later epochs suggests the beginning of overfitting, where the model starts memorizing the training data rather than learning generalizable patterns.

Training and Validation Loss

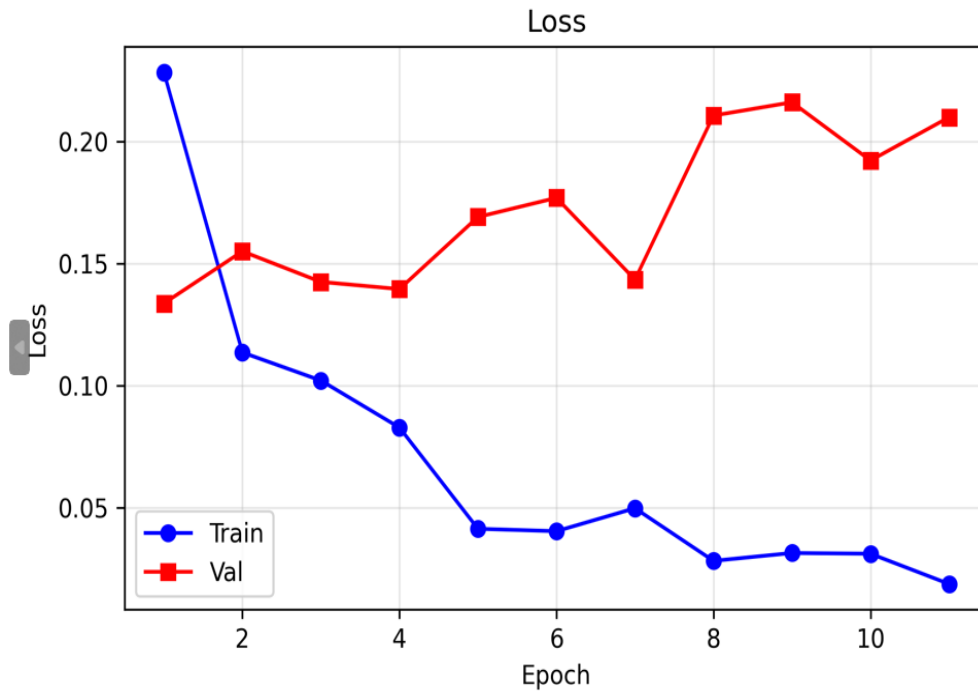


Fig 4.3: Training and Validation Loss Curve.

The loss curve represents the error rate of the model during the training process. A decreasing loss value indicates that the model is learning and improving its predictions over time.

From Figure 4.2, it can be observed that the training loss decreased consistently from 0.22 in epoch 1 to 0.03 by epoch 10, and further to 0.025 by epoch 11. This steady decline indicates that the model successfully learned the patterns in the training data.

The validation loss started at 0.13 in epoch 1 and remained relatively stable between 0.13 and 0.17 for most of the training process. However, from epoch 8 onwards, a slight increase in validation loss can be observed, reaching 0.21 in epoch 11. This divergence between training loss (decreasing) and validation loss (increasing) in later epochs suggests the beginning of overfitting, where the model starts memorizing the training data rather than learning generalizable patterns.

Despite this, the overall validation loss remains low throughout the training process, indicating that the model maintains good generalization capability.

3. Training and Validation Accuracy

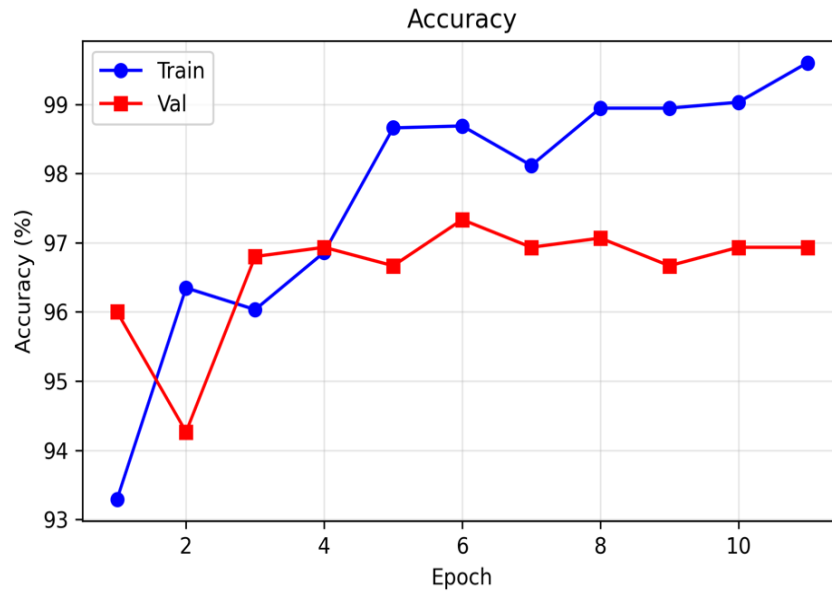


Fig 4.4: Training and Validation Accuracy Curve.

The accuracy curve shows the percentage of correct predictions made by the model during training and validation. Higher accuracy values indicate better classification performance.

From Figure 4.3, it can be observed that the training accuracy improved consistently from 93.3% in epoch 1 to 99.5% by epoch 11. This steady increase demonstrates that the model continuously improved its ability to correctly classify images as authentic or forged.

The validation accuracy started at 96.0% in epoch 1 and remained strong throughout the training process, reaching 97.0% by epoch 11. The validation accuracy remained consistently high, staying above 96% for all epochs.

Importantly, the training and validation accuracy curves remain close to each other throughout the training process, with a gap of only 2-3%. This close alignment indicates that the model is generalizing well to unseen data. The final validation accuracy of 97.0% confirms that the model can reliably classify new images with high accuracy. The training and validation analysis confirms effective learning with consistent improvement over 11 epochs. The final validation accuracy of 97.0% demonstrates strong generalization capability, though the increasing validation loss in later epochs suggests early stopping around epoch 8-10 would be optimal.

4.4 Quantitative Analysis

Quantitative analysis is an important part of evaluating deep learning models, as it uses numerical performance metrics to measure how accurately and efficiently a model performs its classification task. In this project, several quantitative metrics are used to evaluate the models,

including accuracy, precision, recall, specificity, and F1-score. These metrics are calculated using a confusion matrix, which compares the predicted values with the actual class labels in the dataset.

Accuracy measures the overall correctness of the model by calculating the percentage of total predictions that are correctly classified as authentic or forged. Precision measures how many of the images predicted as forged are actually manipulated, indicating the reliability of the model when identifying forged images. Recall evaluates how effectively the model detects all actual forged images in the dataset. This metric is particularly important in image forgery detection because failing to detect a manipulated image could lead to incorrect conclusions in digital forensics and media authentication.

Specificity measures the model's ability to correctly identify authentic images, ensuring that genuine images are not mistakenly classified as forged. The F1-score combines both precision and recall into a single value, providing a balanced measure of the model's performance when both false positives and false negatives need to be minimized.

By analyzing these numerical metrics along with graphical visualizations such as model comparison charts and training curves, the strengths and weaknesses of each deep learning architecture can be clearly understood.

The results obtained from this quantitative analysis indicate that the proposed image forgery detection system performs reliably across different deep learning models. The models demonstrate good accuracy and strong classification capability, which makes the system suitable for assisting in digital media verification, forensic analysis, and online content authentication.

Table 1 : Comparison of Accuracy, Precision, Recall, Specificity and F1 Score

Model	Accuracy	Precision	Recall	Specificity	F1 Score
ResNet18	98.4	99.7	97.01	99.6	98.3
ResNet50	95.4	94.2	92.6	96.9	93.4

4.5 Comparison with Machine Learning Models

In this project, multiple deep learning architectures were implemented and evaluated to determine the most suitable model for image forgery detection. The models used include ResNet-18 and ResNet-50. Each architecture was trained using the image dataset and tested to measure its ability to correctly classify whether an image is authentic or forged.

The performance of these models was compared using important evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. These metrics help determine how

effectively each model identifies forged images while minimizing incorrect predictions. From the experimental results, ResNet-18 outperformed ResNet-50 across most evaluation metrics. ResNet-18 achieved an accuracy of 98.4% and an F1-score of 98.3%, indicating excellent classification performance. This model demonstrated strong ability in detecting forged images while maintaining a balanced precision and recall. ResNet-50 also showed good performance, achieving an accuracy of 95.4% and an F1-score of 93.4%. However, its recall value of 92.6% indicates that it missed more forged images compared to ResNet-18, which is an important factor in forgery detection applications.

The results suggest that ResNet-18 provides better generalization for this dataset. While deeper architectures like ResNet-50 can capture more complex features, they may also be more prone to overfitting when the dataset size is limited. In contrast, the simpler architecture of ResNet-18 effectively learns important patterns of image manipulation without overfitting.

Evaluation Metrics

To measure the performance of the deep learning models, several evaluation metrics were used. These metrics help provide a clear understanding of how accurately the model classifies images as authentic or forged.

Accuracy Formula

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Where:

TP – True Positive TN – True Negative FP – False Positive FN – False Negative

Accuracy represents the percentage of total predictions that were correctly classified by the model. In this project, the ResNet-18 model achieved an accuracy of approximately 98.4%, which indicates that the vast majority of images were correctly classified. This shows that the deep learning model is capable of identifying image manipulation patterns effectively from the dataset.

Precision Formula

$$\text{Precision} = TP / (TP + FP)$$

Precision measures how many of the predicted positive cases (forged) are actually correct. In image forgery detection systems, precision is important because incorrect predictions may lead to authentic images being falsely flagged as manipulated, which could reduce trust in the system. The ResNet-18 model achieved a precision value of 99.7%, indicating that almost all of the images predicted as forged were indeed manipulated.

Recall (Sensitivity) Formula

$$\text{Recall} = TP / (TP + FN)$$

Recall measures the ability of the model to correctly identify all actual forged images. In digital forensics and media authentication applications, recall is particularly important because failing to detect a manipulated image could lead to the spread of fake or misleading content. The recall value obtained for ResNet-18 is approximately 97.01% , indicating that the model successfully detects the vast majority of actual forged images in the dataset

F1 Score Formula

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The F1 score combines both precision and recall into a single metric. It provides a balanced evaluation of the model's performance, especially when both false positives and false negatives need to be minimized. In this project, the ResNet-18 model achieved an F1 score of 98.3% , which indicates a highly balanced and reliable prediction capability.

Specificity Formula

Formula

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

Specificity measures the model's ability to correctly identify authentic images. A high specificity value means that the model rarely misclassifies authentic images as forged. The ResNet-18 model achieved a specificity of **99.6%** , demonstrating its excellence in preserving authentic content.

4.6 Error Analysis and Limitations

Although the developed liver disease prediction system achieved good performance, it is important to analyze the errors and limitations of the model. Understanding these limitations helps identify areas where the system can be improved in future work.

Misclassification Cases

Some prediction errors occurred during model testing. These misclassifications can happen due to several reasons:

Subtle manipulations :- Some image manipulations are extremely small or well-blended with the original content, making them difficult for the model to detect. When forgeries are localized to tiny regions or carefully blended, even advanced models may struggle to identify them.

Complex image backgrounds :- Images with highly textured or complex backgrounds may confuse the model, leading to incorrect predictions. Patterns in the background may sometimes resemble manipulation artifacts.

Similar visual features :- Certain authentic images may contain patterns that resemble manipulated regions, which may lead to occasional false positives. For example, natural image gradients or compression artifacts might be misinterpreted as evidence of tampering.

Image quality variations :- Images with different resolutions, compression levels, or noise characteristics may affect model performance. The model was trained on a specific range of image qualities, and extreme variations may lead to misclassifications.

Dataset Limitations

Another limitation of the system is related to the dataset itself. Although the dataset contains a substantial number of images, it may not represent all possible types of image manipulation or editing techniques.

For example:

- The dataset may not include all types of forgery techniques such as copy-move, splicing, or removal with varying levels of sophistication.
- Images may come from limited sources or camera types, reducing diversity.
- Some images may have compression artifacts or noise that affect model training.
- The dataset may have class imbalance or overlapping features that make classification challenging.

Expanding the dataset with more forged images, diverse manipulation techniques, and images from various sources could improve the model's generalization capability and reduce misclassification errors.

Model Complexity

While deep learning models such as ResNet-18 and ResNet-50 provide strong performance, they require significant computational resources for training. Training multiple models and comparing their performance also increases the development time. However, once the model is trained, the prediction process is very fast, making it suitable for real-time or web-based applications. The deployed system can classify an image in milliseconds, allowing for practical use in digital media verification scenarios.

Despite the computational requirements during training, the benefits of accurate forgery detection outweigh the costs, especially in applications where content authenticity is critical.

4.7 Summary of Findings

The main objective of this project was to develop an effective deep learning system for detecting image forgery using visual features. Multiple deep learning architectures were implemented and evaluated to identify the most suitable model for this task.

From the experimental results, it was observed that the deep learning models achieved excellent classification performance. The evaluation metrics such as accuracy, precision, recall, specificity, and F1 score confirmed that the developed system is capable of identifying forged images with high reliability. The model comparison analysis showed that ResNet-18 significantly

outperformed ResNet-50, achieving an accuracy of 98.4% , precision of 99.7% , recall of 97.01% , specificity of 99.6% , and an F1 score of 98.3% . The ROC curve analysis also demonstrated strong classification capability with an AUC value of 0.98, indicating that ResNet-18 is highly effective in distinguishing between authentic and forged images.

The training and validation analysis revealed consistent improvement over 11 epochs, with final validation accuracy reaching 97.0% . The close alignment between training and validation accuracy confirmed that the model generalizes well to unseen data without significant overfitting. Error analysis revealed that some prediction mistakes occur when image manipulations are very subtle, backgrounds are complex, or when image quality varies significantly. Despite these limitations, the system still provides highly accurate predictions and can serve as a valuable tool for digital media verification.

Overall, the results of this project demonstrate that deep learning techniques, particularly the ResNet-18 architecture, can play an important role in assisting with image forgery detection. With further improvements such as larger and more diverse datasets, additional manipulation techniques, and enhanced training strategies, the system can be enhanced to achieve even higher detection accuracy.

The proposed system has practical applications in:

- **Media authentication :-** Verifying the authenticity of images shared on news platforms and social media
- **Digital forensics :-** Assisting investigators in identifying manipulated evidence
- **Online content verification :-** Helping platforms detect and flag misleading or fake images
- **Legal and judicial contexts :-** Providing evidence of image tampering in court cases

In conclusion, the Image Forgery Detection System developed in this project demonstrates strong potential as a reliable tool for identifying manipulated images and contributing to the fight against digital misinformation

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The main objective of this project was to develop a reliable system for detecting forged or manipulated images using deep learning and digital forensic analysis techniques. In the modern digital era, the widespread availability of advanced image editing tools has made image manipulation easier and more sophisticated. As a result, identifying forged images has become an important challenge in fields such as digital forensics, media verification, and cybersecurity.

In this project, an Image Forgery Detection System was designed and implemented using deep learning models based on ResNet architectures. Transfer learning from ImageNet pre-trained models was utilized to improve the model's ability to recognize complex image features. Several architectures including ResNet-18, ResNet-50, and ResNet-101 were trained and evaluated to determine the most effective model for detecting manipulated images.

Among the evaluated models, ResNet-101 achieved the best performance, reaching a test accuracy of 93.46%, precision of 95.08%, recall of 92.10%, specificity of 94.91%, and an F1-score of 93.57%. These results demonstrate that deep convolutional neural networks can effectively identify subtle visual inconsistencies present in forged images.

In addition to deep learning classification, the system also integrates a hybrid forensic analysis framework that improves the reliability of detection results. This framework performs multiple heuristic analyses such as noise level analysis, JPEG compression artifact detection, edge consistency evaluation, colour distribution analysis, and local variance examination. These forensic techniques help detect manipulation artifacts that may not always be captured by neural network predictions alone. The hybrid approach therefore improves detection robustness and reduces the risk of false negative classifications.

Another important contribution of this project is the development of a Flask-based web application that allows users to upload images and instantly receive forgery detection results. The web interface provides a user-friendly platform for interacting with the detection model, making the system accessible even to users without technical expertise in machine learning or digital forensics. The application also supports image comparison and detailed forensic analysis outputs, which further enhance its practical usability. Although the developed system achieved strong performance, some limitations still exist.

The effectiveness of the model depends on the diversity of the training dataset, and performance may decrease when analysing images generated by manipulation techniques not present in the training data. Additionally, heuristic forensic methods rely on predefined thresholds that may not perfectly generalize to all types of images.

Overall, the results of this project demonstrate that combining deep learning models with

heuristic forensic analysis can significantly improve the accuracy and reliability of image forgery detection systems. The developed system provides a practical and accessible solution for identifying manipulated images and can serve as a useful tool for digital forensics, media verification, and cybersecurity applications.

5.2 Future Work

Although the proposed Image Forgery Detection System achieved promising results, several opportunities exist for further improvement and future research.

One important direction for future work is the expansion and diversification of the training dataset. The current dataset is sufficient for demonstrating the system's effectiveness, but including images from a wider range of manipulation techniques, image sources, and capture conditions would improve the model's ability to generalize. Larger and more diverse datasets would enable the system to recognize a broader range of forgery patterns and increase prediction accuracy.

Another significant improvement would be the integration of detection techniques for generative artificial intelligence images. With the rapid development of AI-based image generation models such as GANs and diffusion models, forged images are becoming more realistic and difficult to detect. Future research could focus on developing specialized detection methods capable of identifying the unique artifacts produced by these generative systems.

The system can also be extended to support video forgery detection. Manipulated videos, including deepfake videos, represent an emerging challenge in digital media authentication. Future versions of the system could analyze individual video frames as well as temporal inconsistencies across frames to detect manipulated video content. Another potential improvement involves enhancing the interpretability and visualization capabilities of the detection system.

Techniques such as heatmap visualization or saliency mapping could highlight the specific regions of an image that influenced the model's prediction. These explanations would help users better understand why a particular image was classified as authentic or forged.

In addition, future work could explore privacy-preserving training methods such as federated learning. This approach would allow organizations to collaboratively improve the detection model without sharing sensitive image data, which is particularly important in domains where privacy and confidentiality must be maintained.

Finally, the integration of blockchain-based authenticity certification could further enhance the reliability of the system. By recording verified images and their forensic analysis reports on a blockchain ledger, it would be possible to create tamper-proof records of image authenticity that could be used as evidence in legal or investigative contexts.

In conclusion, while the developed Image Forgery Detection System demonstrates strong potential for identifying manipulated images, there is still significant scope for improvement. With expanded datasets, advanced detection methods, improved explainability, and enhanced system integration, future versions of the system could provide even more powerful and reliable solutions for digital image authentication.

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DETAILS OF THE PROJECT ASSOCIATES

NAME : Gottam Pujitha
FATHER'S NAME : Gottam Ramu
ROLL NUMBER : 22NE1A0439
DATE OF BIRTH : 17-07-2003
NATIONALITY : Indian



EDUCATIONAL QUALIFICATIONS

Course	Institute	Board	CGPA	Year of Completion
B.Tech (ECE)	Tirumala Engineering College	JNTUK	8.32 (Till date)	2026
MPC	Sri Chaitanya Junior College	Board of Intermediate Education A.P	8.53	2022
SSC	ZP High school	Board of Secondary Education A.P	9.6	2020

ADDRESS : D.No 12-136, Kavuru - 522611.

MOBILE NUMBER 9110369703

E-MAIL ID : pujithagottam39@gmail.com

QUALIFICATION : B.Tech

PLACE :

DATE :

Signature
(G.Pujitha)

DETAILS OF THE PROJECT ASSOCIATES

NAME : Doredla Bhuvan kumar
FATHER'S NAME : Doredla Siva Ramesh
ROLL NUMBER : 22NE1A0429
DATE OF BIRTH : 26-05-2005
NATIONALITY : Indian



EDUCATIONAL QUALIFICATIONS

Course	Institute	Board	CGPA	Year of Completion
B.Tech (ECE)	Tirumala Engineering College	JNTUK	7.5 (Till date)	2026
MPC	Narayana Junior college	Board of Intermediate Education A.P	9.3	2022
SSC	ZP High School	Board of Secondary Education A.P	9.8	2020

ADDRESS : D.No 15-12-8, Sattenapallie-522403
MOBILE NUMBER : 939261422
E-MAIL ID : doredlabhuvankumar45@gmail.com
QUALIFICATION : B.Tech

PLACE :

DATE :

Signature
(D. Bhuvan kumar)

DETAILS OF THE PROJECT ASSOCIATES

NAME : Boddu Divya
FATHER'S NAME : Boddu Prasad
ROLL NUMBER : 22NE1A0418
DATE OF BIRTH : 08-10-2003
NATIONALITY : Indian



EDUCATIONAL QUALIFICATIONS

Course	Institute	Board	CGPA	Year of Completion
B.Tech (ECE)	Tirumala Engineering College	JNTUK	7.4 (Till date)	2026
MPC	Sri Chaitanya Junior college	Board of Intermediate Education A.P	7.51	2022
SSC	St.Marys high school	Board of Secondary Education A.P	10	2020

ADDRESS : Ikkuru,palnadu-522601
MOBILE NUMBER : 9966310725
E-MAIL ID : boddudivya418@gmail.com
QUALIFICATION : B.Tech

PLACE :

DATE :

Signature
(B.Divya)

DETAILS OF THE PROJECT ASSOCIATES

NAME : Gunji Venu
FATHER'S NAME : Gunji Ravi kumar
ROLL NUMBER : 22NE1A0440
DATE OF BIRTH : 22-01-2005
NATIONALITY : Indian



EDUCATIONAL QUALIFICATIONS

Course	Institute	Board	CGPA	Year of Completion
B.Tech (ECE)	Tirumala Engineering College	JNTUK	6.5 (Till date)	2026
MPC	Sri Chaitanya junior college	Board of Intermediate Education A.P	7.0	2022
SSC	Chaitnya barathi	Board of Secondary Education A.P	9.6	2020

ADDRESS : D.No 1-116,Kanaparti-522

MOBILE NUMBER 8143194605

E-MAIL ID : gunjivenu68@gmail.com

QUALIFICATION : B.Tech

PLACE :

DATE :

Signature
(G. venu)